A Cluster Analysis for Discovering the Psychosocial Profiles of Mathematically Resilient Students in Latin America

Abstract. Educational inequality in Latin America coexists with a less discussed phenomenon: students who achieve high mathematics performance despite disadvantaged socioeconomic backgrounds. This study examines whether these students form a homogeneous group or fall into distinct psychosocial profiles. We analyzed 713 mathematically resilient students from six Latin American countries (Argentina, Brazil, Chile, Dominican Republic, Panama, and Uruguay) participating in PISA 2022, using 12 motivational, socioemotional, and contextual indices. Ward.D2 hierarchical clustering was combined with the K-Means algorithm, validated by average silhouette width. Three profiles emerged: High Motivation and Support (29%), Self-Directed (32%), and High Anxiety (39%). Mathematical resilience is not a unitary phenomenon; it manifests through distinct configurations of protective factors, with direct implications for differentiated pedagogical interventions.

Keywords: mathematical resilience; cluster analysis; PISA 2022; Latin America; psychosocial profiles.

1. Introduction

Educational inequality is one of the most persistent challenges facing Latin American societies. The 2022 results of the Programme for International Student Assessment (PISA) once again reveal that students from low socioeconomic status families tend to perform substantially worse in mathematics than their peers in more advantaged contexts, perpetuating cycles of exclusion that undermine both human development and long-term economic performance (OECD, 2023b). Nevertheless, a subset of these students manages to overcome this reality and achieve high performance, a phenomenon described in the literature as academic resilience (Masten, 2001) and, in the specific domain of mathematics, one that has attracted increasing attention as an object of empirical investigation (Cheung et al., 2024; Johnston-Wilder and Lee, 2010).

Johnston-Wilder and Lee (2010) introduced the concept of mathematical resilience as a positive affective stance toward the inherent challenges of mathematical reasoning, one that involves persisting, seeking help when necessary, and learning from mistakes. In PISA, the construct acquires a distributive character: students are considered resilient if, despite belonging to the lowest quartile of the economic, social, and cultural status index (ESCS) within their own country, they attain scores in the highest performance quartile within the same economy (OECD, 2023b). This definition contextualizes the criterion relative to national distributions, enabling comparisons across structurally distinct educational systems, an indispensable condition for regional analyses such as the one proposed here.

The literature has accumulated evidence on protective factors associated with mathematical resilience across multiple systemic levels. At the individual level, mathematical self-efficacy emerges as the strongest predictor (Niu, Xu, and Yu, 2025); growth mindset moderates intrinsic motivation and is associated with lower anxiety (Dong, Jia, and Fei, 2023); and mathematical anxiety, the primary individual risk factor, is mitigated by the emotional support of teachers (Tan et al., 2024). At the family level, parental involvement strengthens self-efficacy (Djekourmane et al., 2025). At the school level, disciplinary climate, teacher quality, and the mean socioeconomic composition of the school emerge as the most consistent institutional predictors (Agasisti et al., 2018; Cheung et al., 2024).

Despite the growing body of evidence, research in this field carries an important limitation: the predominance of variable-centered approaches that estimate average effects across the population and thereby obscure the internal heterogeneity of resilient students (Ye, Teig, and Blömeke, 2024). Multilevel models and regression analyses effectively describe what predicts resilience, but say little about who resilient students are and which specific combinations of protective factors characterize subgroups with distinct trajectories. This gap is particularly salient in the Latin American context, where differences among national educational systems, in terms of funding, teacher training, curricular coverage, and family social capital, make it unlikely that a single resilience pattern would adequately describe the region as a whole.

Cluster analysis techniques offer a complementary methodological alternative particularly well suited to this purpose. By organizing observations into internally homogeneous and mutually distinct groups without relying on predefined categorical labels, these methods can identify latent co-occurrence patterns of psychosocial, motivational, and contextual attributes that mean-based analyses tend to mask (Jain, 2010). Recent applications to PISA 2022 have already demonstrated the potential of this approach for revealing profiles with distinct psychosocial configurations (Alvarez-Garcia, Arenas-Parra, and Ibar-Alonso, 2024). However, the application of these techniques to the specific subset of mathematically resilient students in Latin America remains underexplored in the available literature.

It is within this context that the present study is situated. Cluster analysis was applied to mathematically resilient students identified in PISA 2022 across six Latin American countries, Argentina, Brazil, Chile, the Dominican Republic, Panama, and Uruguay, combining Ward.D2 hierarchical clustering with the K-Means algorithm and validating the solution by average silhouette width. The central objective was to identify and characterize psychosocially distinct profiles of mathematical resilience. The contribution operates on two dimensions: methodological, through the application of a person-centered approach to an analytical subset rarely treated as an object of internal segmentation; and substantive, by demonstrating that mathematical resilience is not a unitary phenomenon in the region, but rather manifests through structurally distinct configurations of protective factors, each with specific implications for educational policy.

2. Theoretical Background

2.1 Mathematical Resilience

The concept of mathematical resilience was formulated by Johnston-Wilder and Lee (2010) as a positive affective stance of the learner toward the challenges inherent to mathematics learning. The original definition describes the capacity to engage, persist, and make use of mathematics both inside and outside school, even in the face of significant and prolonged obstacles. It is not, therefore, a fixed trait, but rather a cultivable disposition modulated by affective, cognitive, and relational factors.

In the PISA context, academic resilience is defined in operational terms: resilient students are those who, despite belonging to the lowest 25% of the economic, social, and cultural status index (ESCS) within their country, attain scores in the highest 25% of performance within the same economy (OECD, 2023b). This was the definition adopted in the present study, applied specifically to mathematics proficiency.

2.2 Factors Associated with Mathematical Resilience

Factors associated with mathematical resilience are distributed across multiple levels, individual, family, and school, and operate interdependently on the student's capacity to persist and succeed in mathematics despite socioeconomic adversity (Ye, Teig, and Blömeke, 2024). At the individual level, mathematical self-efficacy is the most robust predictor: students who believe in their own abilities tend to set more ambitious goals, invest greater effort, and persist in the face of difficulties, with a protective effect that remains significant even after controlling for socioeconomic status in multilevel analyses with PISA data (Niu, Xu, and Yu, 2025). Growth mindset acts as a moderator of intrinsic motivation and engagement with challenging tasks, and is associated with lower mathematical anxiety (Dong, Jia, and Fei, 2023). Mathematical anxiety, the primary individual risk factor, is strongly mitigated by the perception of emotional support from teachers (Tan et al., 2024).

At the family level, parental involvement strengthens self-efficacy and growth mindset, reduces avoidance, and promotes perseverance in mathematics over time (Djekourmane et al., 2025). At the school level, a favorable disciplinary climate, teacher quality, and the mean socioeconomic composition of the school emerge as the most consistent institutional predictors of mathematical resilience in large-scale assessments, operating as compensatory mechanisms for students from disadvantaged contexts (Agasisti et al., 2018; Cheung et al., 2024).

2.3 Cluster Analysis

Cluster analysis is one of the fundamental approaches to unsupervised learning. Its objective is to organize a set of objects into internally homogeneous and mutually distinct groups without the use of predefined categorical labels (Jain, 2010). Formally, clustering seeks a partition of the data in which observations within the same group share greater similarity among themselves than with observations from other groups, as measured by distance metrics applied to the attribute space (Xu and Tian, 2015). The main paradigms include partitioning methods, of which K-Means is the most widely used; hierarchical methods, which produce dendrogram-like structures without requiring the number of groups to be specified in advance; density-based methods; and model-based methods, such as Gaussian mixtures and Latent Profile Analysis (LPA), which treat group membership as a probabilistic latent variable (Xu and Wunsch, 2005). The choice of algorithm, distance metric, and number of groups must be guided by internal validation criteria, such as silhouette indices, the Davies-Bouldin index, and BIC, and by the substantive interpretability of the resulting clusters (Xu and Tian, 2015). In educational data science, clustering has been employed to identify latent student profiles from cognitive, affective, and socioeconomic indicators, revealing heterogeneous patterns that mean-centered analyses typically obscure (Alvarez-Garcia, Arenas-Parra, and Ibar-Alonso, 2024).

3. Material and Methods

3.1 Study Design

The study adopts a quantitative, cross-sectional, and descriptive-exploratory design based on secondary data from the 2022 Programme for International Student Assessment (PISA), conducted by the Organisation for Economic Co-operation and Development (OECD). The analysis focuses on a subset of Latin American countries and employs unsupervised clustering techniques to identify profiles among students classified as mathematically resilient. The approach follows the exploratory framework proposed by Oliveira et al. (2022), combining hierarchical and non-hierarchical methods validated by internal clustering quality indices.

3.2 Data Source and Sample

The data derive from the public PISA 2022 database, made available by the OECD in SPSS format (.sav). The analytical sample comprises students with records of participation from six Latin American countries: Argentina (ARG), Brazil (BRA), Chile (CHL), Dominican Republic (DOM), Panama (PAN), and Uruguay (URY). The inclusion of these six countries was guided by two criteria: complete data availability for the variables of interest, and regional representativeness of Latin America in the 2022 cycle. From the total universe of students in the selected countries, the subset classified as mathematically resilient (see Section 3.3) was extracted.

3.3 Operationalization of Mathematical Resilience

Mathematical resilience was operationalized through a bivariate distributional criterion consistent with the established approach in PISA literature (OECD, 2023a). A student was classified as mathematically resilient (resil_math = 1) when simultaneously satisfying the following conditions: (i) belonging to the lowest quartile of the economic, social, and cultural status index (ESCS) distribution within their country of residence; and (ii) attaining a mathematics score in the highest performance quartile of their country's distribution, estimated through Plausible Values (PVs).

This criterion ensures that the resilience construct is defined in relative terms, that is, relative to the internal distribution of each country, thereby preventing systemic differences among national educational systems from distorting the classification. The variable resil_math was computed prior to the present analysis and is already integrated into the processed dataset.

3.4 Clustering Variables

The set of variables used in the cluster analysis comprised 12 psychosocial, motivational, and contextual indices derived by the PISA consortium through Weighted Likelihood Estimates (WLE). These indices result from Item Response Theory (IRT) models applied to self-report scales in the student and school questionnaires, and possess international standardization that ensures comparability across countries and cycles.

The criterion for variable inclusion was twofold: (a) less than 30% missing data in the resilient student subset; and (b) theoretical relevance to the motivational, socioemotional, and contextual domains associated with academic resilience. Table 1 presents the selected variables, organized by analytical dimension.

Table 1 – Variables used in the cluster analysis, by analytical dimension

Code	Dimension	Description
MATHEFF	Motivation	Mathematical self-efficacy
ANXMAT	Motivation	Mathematical anxiety (higher scores = greater anxiety)
MATHPERS	Motivation	Effort and persistence in mathematics
CURIOAGR	Personality	Intellectual curiosity
EMOCOAGR	Personality	Emotional control
GROSAGR	Personality	Growth mindset
TEACHSUP	School context	Perceived support from the mathematics teacher
DISCLIM	School context	Disciplinary climate in the mathematics classroom
BELONG	Well-being	Sense of school belonging
COGACRCO	Pedagogical practice	Cognitive activation – fostering reasoning
EXPOFA	Curricular exposure	Exposure to formal and applied mathematical tasks

EXPO21ST	Curricular exposure	Exposure to reasoning tasks and 21st-century competencies

Note. All indices are expressed on the WLE scale standardized by the PISA 2022 consortium.

3.5 Data Preparation

3.5.1 Missing data treatment

A complete-case analysis strategy (listwise deletion) was adopted, retaining only resilient students with valid records on all 12 clustering variables. The proportion of retained cases was verified and reported as an indicator of potential selection bias. Although multiple imputation represents a methodologically more conservative alternative, complete-case analysis was preferred given the exploratory nature of the study and the absence of evidence of a systematic missingness pattern in this specific subset.

3.5.2 Variable standardization

Although WLE variables are, by construction, comparable across countries and dimensions, their internal range may vary considerably within the resilient student subset. To prevent any single variable from exerting disproportionate influence on the distance metric used in clustering, all 12 variables were re-standardized to z-scores (mean zero and unit standard deviation) specifically within this subset. The resulting standardized matrix, denoted X, constituted the direct input to the clustering algorithms. Descriptive statistics on the original WLE scale (mean, standard deviation, minimum, and maximum) were calculated and reported for purposes of substantive profile interpretation.

3.6 Determining the Number of Clusters

The determination of the optimal number of groups (k) was conducted through a dual strategy combining hierarchical and non-hierarchical approaches, validated by an internal clustering quality index.

A Euclidean distance matrix was computed on the standardized matrix X, and agglomerative hierarchical clustering was performed using Ward's linkage method (Ward.D2 variant) (Murtagh and Legendre, 2014), implemented via the hclust() function in R. The Ward.D2 method minimizes the within-group sum of squares at each merging step, making it particularly suitable for continuous variables on a common scale and tending to produce compact, balanced-size clusters. The resulting dendrogram was visually inspected to identify the number of groups suggested by the merging structure (elbow criterion).

For each value of k in the range [2, 6], two models were fitted: (a) K-Means with 50 random initializations (nstart = 50) and a maximum of 100 iterations; and (b) a cut of the Ward hierarchical dendrogram at the level corresponding to k groups. In both cases, the average silhouette width (sw) was calculated, defined by the equation:

$$\overline{S} = \frac{1}{n}\sum_{i=1}^{n} \frac{b(i)-a(i)}{max\{a(i),b(i)\}}$$

where a(i) is the mean Euclidean distance from individual i to all other members of its own cluster, and b(i) is the smallest mean Euclidean distance from individual i to members of any other cluster. Values of sw close to 1 indicate high internal cohesion and good inter-cluster separation; values near 0 suggest overlap; negative values indicate incorrect cluster assignment. Results were plotted in a comparative line chart for both methods and tabulated for k = 2 to 6.

The number of clusters k = 3 was selected based on three convergent criteria, summarized in Table 2.

Table 2 – Convergent criteria for the selection of k = 3

Criterion	Result	Role
Visual inspection of dendrogram	Prominent elbow suggesting k = 3	Corroborative
Average silhouette – K-Means	Maximum at k = 3 in the range [2, 6]	Decisive
Average silhouette – Ward.D2	Consistent with K-Means at k = 3	Confirmatory
Theoretical framework	Three recurrent profiles in academic resilience studies (Oliveira et al., 2022)	Confirmatory

The final solution was obtained by running the K-Means algorithm with nstart = 100 random restarts and a maximum of 200 iterations, a more conservative configuration than that used in the exploratory stage, in order to minimize the probability of convergence to a local optimum. The random seed was fixed at 2024 to ensure reproducibility. The average silhouette width and the composition of each cluster were verified in the final solution.

3.7 Profile Characterization and Naming

The identified clusters were characterized through three complementary procedures. First, the means of the 12 WLE variables on the original scale were computed for each cluster, enabling substantive interpretation of the profiles in measurement units comparable to the PISA literature. The mean mathematics score and mean ESCS index per cluster were also reported as contextual verification indicators for the operationalization of resilience. Second, the standardized centroid means (z-scores) from K-Means were represented graphically in a parallel coordinates plot, enabling simultaneous visual comparison of the profile shape across all 12 dimensions. Third, the data were projected onto the two-dimensional space of the first two principal components (Principal Component Analysis), with clusters represented by convex ellipses, to visually assess the geometric separability among the identified profiles. The distribution of students by country and cluster was computed and presented in a proportional bar chart and a cross-tabulation of frequencies.

Profile naming was carried out programmatically based on the variable with the greatest a priori discriminant power (mathematical self-efficacy, MATHEFF), with manual adjustment following inspection of the full set of means. The profiles were designated Profile A (High Motivation and Support), Profile B (Self-Directed), and Profile C (High Anxiety), in descending order of mean self-efficacy and following the predominant configuration observed.

3.8 Cross-Method Validation

To assess the robustness of the K-Means solution, a systematic comparison with the Ward hierarchical solution (k = 3) was conducted at two levels. At the global quality level, the average silhouette widths of both methods and the relative cluster sizes were compared. At the individual agreement level, a contingency matrix was constructed by crossing each student's cluster assignments under the two methods (K-Means × Ward), in order to quantify the degree of consistency in profile identification regardless of the algorithm used. High agreement between methods was taken as evidence of structural stability in the identified clusters.

3.9 Software and Reproducibility

All analyses were conducted in the R environment (version 4.x), with the random seed fixed at 2024 (set.seed(2024)). The following packages were used: haven (reading SPSS files), dplyr and tidyr (data manipulation), cluster (silhouette computation), factoextra (clustering and PCA visualizations), dendextend (dendrogram), and ggplot2 with patchwork (figures).

4. Results and Discussion

4.1 Determining the Number of Clusters

The Ward.D2 dendrogram (Figure 1) revealed a tripartite structure with a natural cut around distance 30, suggesting k = 3 as the visually most coherent solution.

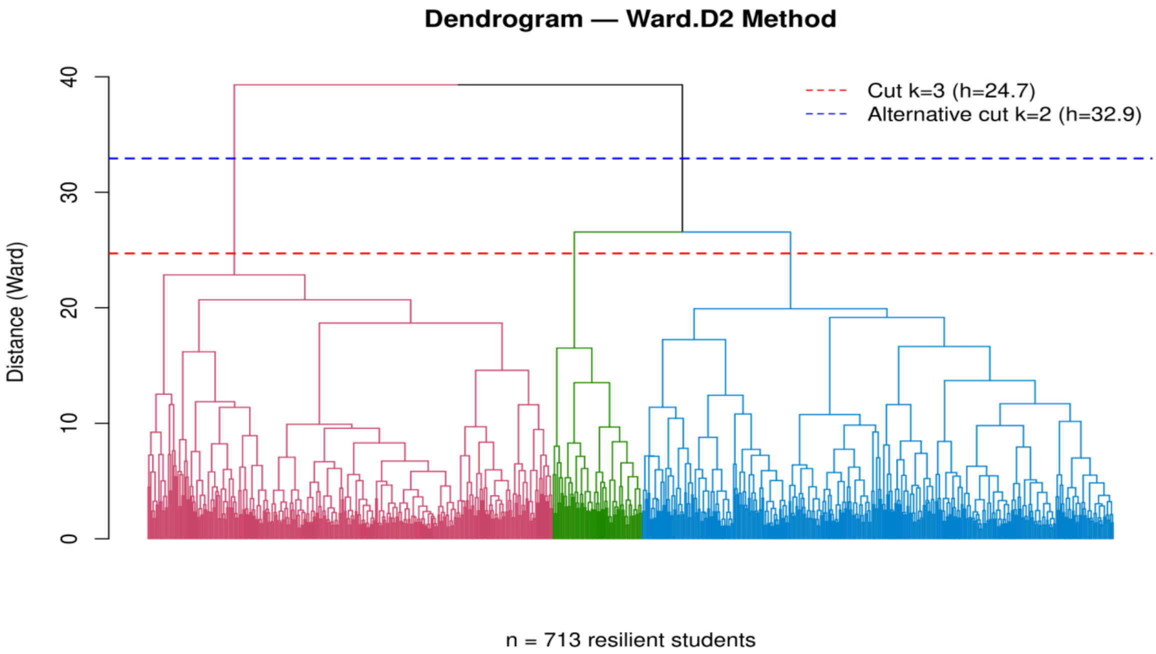

Figure 1 – Ward.D2 hierarchical dendrogram of mathematically resilient students

Table 1 and Figure 2 present the silhouette width for k = 2 to 6 under both methods. K-Means outperformed Ward across all tested configurations. The value k = 3 was selected as the final solution: it yields the second-highest silhouette for K-Means (0.083), but this configuration offers clearly superior substantive interpretability compared to k = 2 and maintains coherence with the literature on resilience profiles (Masten, 2001; Oliveira et al., 2022).

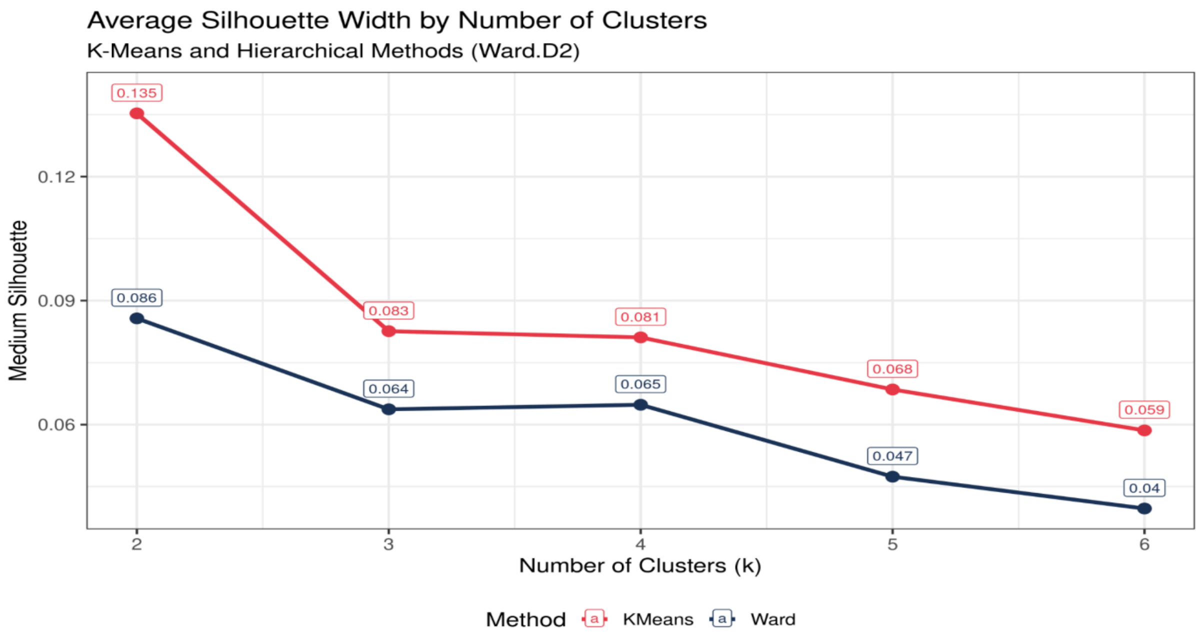

Figure 2 – Average silhouette width by number of clusters and method

4.2 Characterization of the Three Profiles

The K-Means solution with k = 3 produced clusters of sizes 207 (Profile A), 228 (Profile B), and 278 (Profile C), with an overall silhouette of 0.083. Table 2 summarizes the centroid means across the 12 variables on the original WLE scale, while Figure 3 and Figure 4 offer complementary readings through parallel coordinates and radar chart visualizations, respectively.

Table 2 – Variable means by profile (original WLE scale)

Variable	Profile A (n=207)	Profile B (n=228)	Profile C (n=278)
MATHEFF	0.368	−0.781	−0.638
ANXMAT	−0.478	0.311	0.516
MATHPERS	1.100	−0.142	−0.089
CURIOAGR	1.163	0.302	−0.182
EMOCOAGR	0.533	0.028	−0.140
GROSAGR	0.743	0.314	0.117
TEACHSUP	0.951	−0.326	0.561
DISCLIM	0.167	−0.472	−0.269
BELONG	0.475	−0.281	−0.408
COGACRCO	0.618	−0.448	0.227
EXPOFA	0.041	−0.652	0.382
EXPO21ST	0.404	−0.619	0.539
Math score (mean)	475.3	467.6	459.5
ESCS (mean)	−1.970	−2.020	−1.999

Note. Positive values of ANXMAT indicate higher mathematical anxiety. For all other variables, positive values indicate attributes above the mean of the resilient student subset.

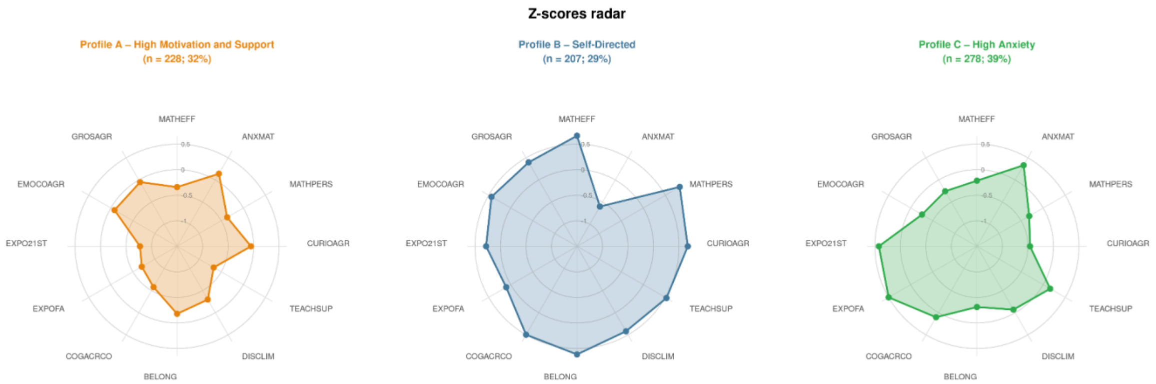

Figure 3 – Centroid profiles by cluster (radar chart)

Profile A – High Motivation and Support (n = 207; 29.0%). This profile groups students with the highest mathematical self-efficacy (MATHEFF = 0.368), the lowest anxiety (ANXMAT = −0.478), elevated persistence (MATHPERS = 1.100), and strong intellectual curiosity (CURIOAGR = 1.163). These students perceive the greatest teacher support (TEACHSUP = 0.951) and report meaningful exposure to cognitively activating pedagogical practices (COGACRCO = 0.618). The mean mathematics score is 475.3 points. These are the resilient students "protected" by a broad and simultaneous set of favorable factors.

Profile B – Self-Directed (n = 228; 32.0%). This profile is distinguished primarily by the lowest exposure to formal and applied tasks (EXPOFA = −0.652) and to 21st-century competencies (EXPO21ST = −0.619), worse perceived disciplinary climate (DISCLIM = −0.472), and lower teacher support (TEACHSUP = −0.326). The mean score is 467.6 points. The resilience of these students appears to be sustained by extracurricular or family factors not fully captured by the model variables.

Profile C – High Anxiety (n = 278; 39.0%). This is the largest group. It presents the highest mathematical anxiety (ANXMAT = 0.516) and a low self-efficacy (MATHEFF = −0.638). In an apparently paradoxical finding, this profile records above-average teacher support (TEACHSUP = 0.561) and strong curricular exposure (EXPOFA = 0.382; EXPO21ST = 0.539), suggesting that institutional support exists but does not translate into mathematical self-confidence. The mean score is 459.5 points. This profile holds the greatest relevance for pedagogical interventions aimed at developing self-efficacy and reducing anxiety (Bandura, 1997).

4.3 Distribution by Country

Table 3 present the profile composition by country. Profile C predominates in Argentina (45%) and Brazil (46%), suggesting that in these countries resilience manifests despite, rather than because of, a disadvantageous combination of low self-efficacy and high anxiety. Uruguay concentrates the highest proportion of students in Profile B (46%), while Chile stands out for the greatest share in Profile A (39%), indicating that its school conditions appear more conducive to the construction of mathematical self-efficacy even among students from low socioeconomic backgrounds.

Table 3 – Absolute and relative distribution of profiles by country

Country	Profile A	Profile B	Profile C	Total
Argentina	36 (21%)	57 (34%)	76 (45%)	169
Brazil	52 (29%)	46 (26%)	82 (46%)	180
Chile	43 (39%)	30 (27%)	38 (34%)	111
Dominican Republic	33 (34%)	27 (28%)	37 (38%)	97
Panama	10 (27%)	13 (35%)	14 (38%)	37
Uruguay	33 (28%)	55 (46%)	31 (26%)	119
Total	**207 (29%)**	**228 (32%)**	**278 (39%)**	**713**

These findings suggest that educational systems may produce distinct forms of resilience, some more sustainable from a socioemotional standpoint than others. The cross-country comparison highlights that academic performance, in isolation, is insufficient for understanding the educational experience of resilient students: it is also necessary to consider the psychological and contextual mechanisms that sustain success in contexts of social vulnerability.

5. Conclusions

The results clearly show that mathematical resilience among Latin American students from low socioeconomic backgrounds participating in PISA 2022 is not a unitary phenomenon. Cluster analysis identified three psychosocially distinct profiles that differ not only in magnitude, but in the very configuration of the protective mechanisms involved. This structural heterogeneity represents a substantive contribution to the field, as it challenges population mean-centered approaches and reinforces the need for person-centered models in the investigation of academic resilience (Masten, 2001; Oliveira et al., 2022).

The three identified profiles suggest distinct pathways to resilience. In Profile A (High Motivation and Support, 29.0%), resilience emerges from the convergence of multiple favorable individual and contextual factors, a pattern consistent with cumulative protection models (Ye, Teig, and Blömeke, 2024). In Profile B (Self-Directed, 32.0%), resilience persists even amid unfavorable school conditions, suggesting that its protective mechanisms reside in domains not captured by PISA's institutional variables, such as family social capital and extracurricular support networks. Profile C (High Anxiety, 39.0%), the most numerous and the one with the greatest interventional relevance, presents a paradox: above-average institutional support does not translate into self-confidence. This is consistent with Bandura (1997), for whom self-efficacy requires mastery experiences interpreted positively by the learner, and points to the centrality of formative feedback and metacognitive strategies in contexts of vulnerability.

The distribution of profiles across countries reveals patterns with implications for comparing educational systems. Chile concentrates the highest proportion of students in Profile A (39%), suggesting more robust school conditions for building self-efficacy among disadvantaged students. Argentina and Brazil, by contrast, show a predominance of Profile C (45% and 46%, respectively), indicating that resilience in these contexts manifests in spite of, rather than thanks to, favorable motivational conditions. Uruguay stands out for its concentration in Profile B (46%), a pattern that warrants specific investigation of the extracurricular mechanisms sustaining these students. Such differences should be read as products of systemic configurations that unequally structure access to protective experiences, rather than as reflections of innate individual capacities.

From the perspective of educational policy, uniform interventions tend to be inefficient precisely because they fail to address the heterogeneity of profiles. For Profile C, structured programs for developing mathematical self-efficacy are needed, incorporating explicit instruction in metacognitive strategies, restructuring of causal attribution beliefs, and systematic cycles of formative feedback. For Profile A, curricular enrichment and advanced cognitive challenges appear more appropriate. Profile B, finally, requires an in-depth diagnosis of extracurricular protective factors before any direct intervention, lest ongoing protective mechanisms be inadvertently disrupted.

Several methodological limitations deserve acknowledgment. The exclusion of 29.7% of cases through complete-case analysis may introduce selection bias. K-Means assumes spherical clusters and homogeneous variance, which may underrepresent irregular-boundary structures. The cross-sectional design precludes causal inference. Furthermore, the silhouette of 0.083 for $k = 3$, while superior to all tested alternatives, signals partial overlap among groups, an expected behavior in high-dimensional psychosocial data that reinforces the exploratory nature of the solution.

Future research may advance along three lines: (i) replacing complete-case analysis with multiple imputation prior to clustering; (ii) applying Latent Profile Analysis, a more rigorous probabilistic alternative for self-report data; and (iii) extending the analysis to all eleven Latin American countries in PISA 2022, recalculating resilience criteria on the expanded sample, in order to assess the generalizability of the identified profiles and explore cross-system variation with greater comparative power.

Acknowledgments

[Acknowledgment text will be here, if aproved.]

References

AGASISTI, T.; AVVISATI, F.; BORGONOVI, F.; LONGOBARDI, S. Academic resilience: what schools and countries do to help disadvantaged students succeed in PISA. OECD Education Working Papers, n. 167. Paris: OECD Publishing, 2018. DOI: 10.1787/e22490ac-en.

ALVAREZ-GARCIA, M.; ARENAS-PARRA, M.; IBAR-ALONSO, R. Uncovering student profiles: an explainable cluster analysis approach to PISA 2022. Computers & Education, v. 223, p. 105166, Dec. 2024. DOI: 10.1016/j.compedu.2024.105166.

BANDURA, A. Self-efficacy: the exercise of control. New York: W. H. Freeman, 1997. ISBN 978-0-7167-2850-4.

CHEUNG, K. C. et al. A machine-learning model of academic resilience in the times of the COVID-19 pandemic: evidence drawn from 79 countries/economies in the PISA 2022 mathematics study. British Journal of Educational Psychology, v. 94, n. 4, p. 1224–1244, Dec. 2024. DOI: 10.1111/bjep.12715.

DJEKOURMANE, D. et al. The mediating role of math self-efficacy between school-based parental involvement and math performance among students in Southeast Asia: evidence from PISA 2022. Frontiers in Psychology, v. 16, p. 1594131, Aug. 2025. DOI: 10.3389/fpsyg.2025.1594131.

DONG, L.; JIA, X.; FEI, Y. How growth mindset influences mathematics achievements: a study of Chinese middle school students. Frontiers in Psychology, v. 14, p. 1148754, Mar. 2023. DOI: 10.3389/fpsyg.2023.1148754.

JAIN, A. K. Data clustering: 50 years beyond K-means. Pattern Recognition Letters, v. 31, n. 8, p. 651–666, Jun. 2010. DOI: 10.1016/j.patrec.2009.09.011.

JOHNSTON-WILDER, S.; LEE, C. Developing mathematical resilience. In: BERA ANNUAL CONFERENCE, 2010, Warwick. Proceedings... Warwick: British Educational Research Association, 2010. Available at: wrap.warwick.ac.uk/30602.

MASTEN, A. S. Ordinary magic: resilience processes in development. American Psychologist, v. 56, n. 3, p. 227–238, 2001. DOI: 10.1037/0003-066X.56.3.227.

MURTAGH, F.; LEGENDRE, P. Ward's hierarchical agglomerative clustering method: which algorithms implement Ward's criterion? Journal of Classification, v. 31, n. 3, p. 274–295, 2014. DOI: 10.1007/s00357-014-9161-z.

NIU, J.; XU, H.; YU, J. Identifying multilevel factors on student mathematics performance for Singapore, Korea, Finland, and Denmark in PISA 2022: considering individualistic versus collectivistic cultures. Humanities and Social Sciences Communications, v. 12, n. 1, p. 151, Feb. 2025. DOI: 10.1057/s41599-025-04466-y.

OECD. PISA 2022 assessment and analytical framework. Paris: OECD Publishing, 2023a. DOI: 10.1787/dfe0bf9c-en.

OECD. PISA 2022 results (Volume I): the state of learning and equity in education. Paris: OECD Publishing, 2023b. DOI: 10.1787/53f23881-en.

OLIVEIRA, P. L. S. et al. Identificação de pesquisas e análise de algoritmos de clusterização para a descoberta de perfis de engajamento. Revista Brasileira de Informática na Educação, v. 30, p. 1–19, 2022. DOI: 10.5753/rbie.2022.2508.

TAN, Y. et al. How teacher-student closeness and conflict contribute to mathematical problem solving in Chinese adolescents: a multilevel moderated mediation model of self-efficacy and school climate. Current Psychology, v. 43, n. 4, p. 3111–3123, Jan. 2024. DOI: 10.1007/s12144-023-04382-x.

XU, D.; TIAN, Y. A comprehensive survey of clustering algorithms. Annals of Data Science, v. 2, n. 2, p. 165–193, Jun. 2015. DOI: 10.1007/s40745-015-0040-1.

XU, R.; WUNSCH, D. Survey of clustering algorithms. IEEE Transactions on Neural Networks, v. 16, n. 3, p. 645–678, May 2005. DOI: 10.1109/TNN.2005.845141.

YE, W.; TEIG, N.; BLÖMEKE, S. Systematic review of protective factors related to academic resilience in children and adolescents: unpacking the interplay of operationalization, data, and research method. Frontiers in Psychology, v. 15, p. 1405786, Aug. 2024. DOI: 10.3389/fpsyg.2024.1405786.